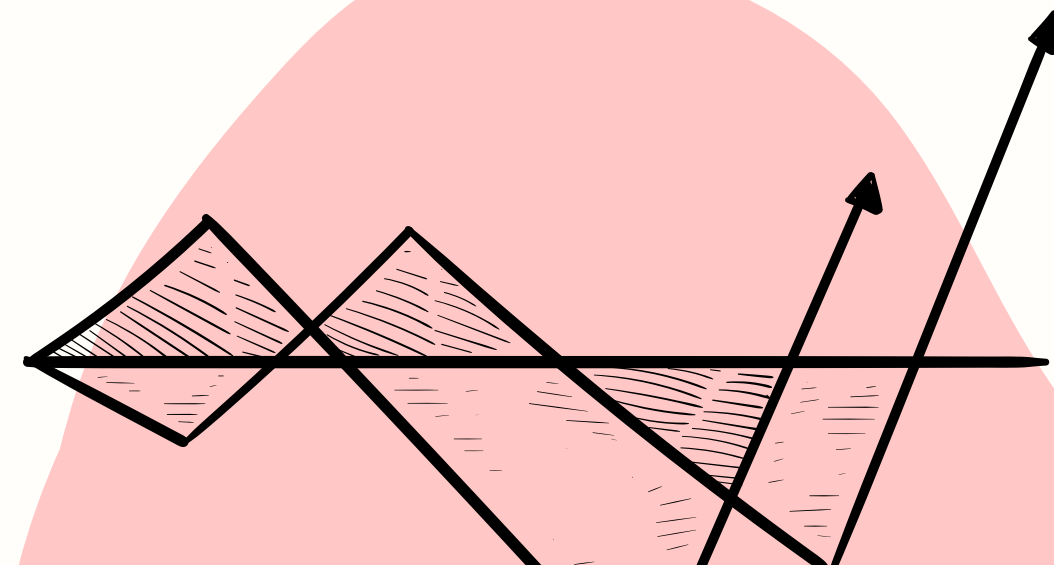
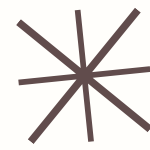
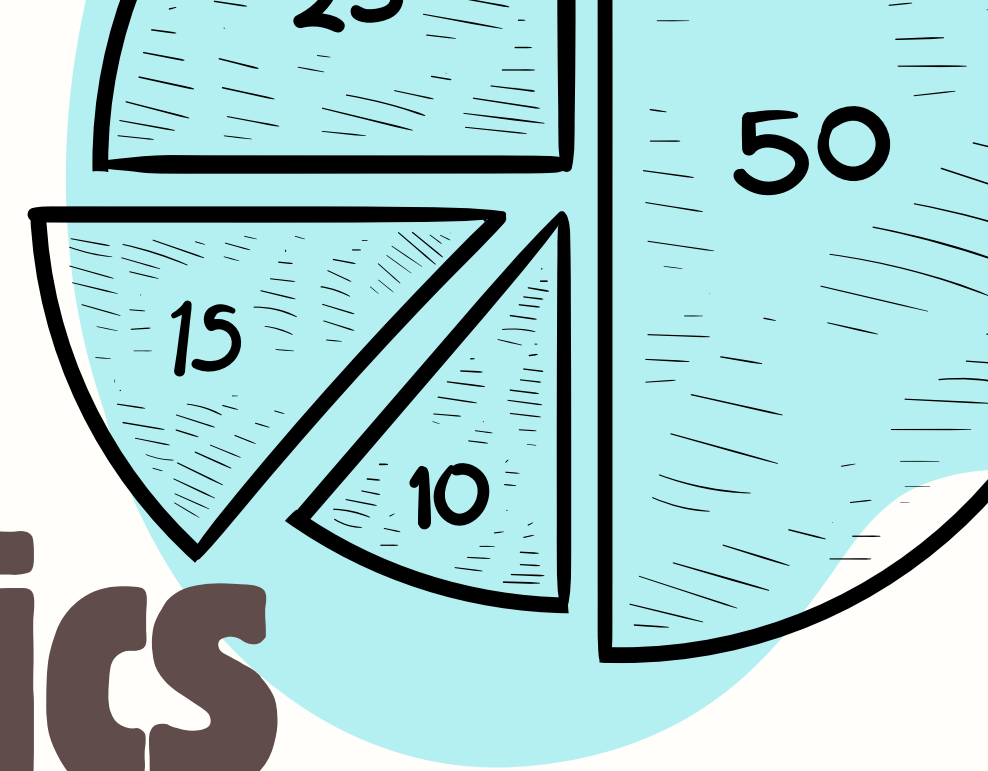


Predictive Analytics for Early Clinical Deterioration



Team

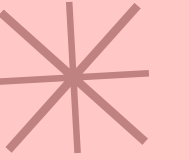
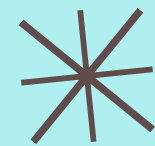
Simran Amesar

Ishwari Thakur

Atharv Sawant

Yuvraj Ghag

Nidhi Nagaraja



Project Overview

✱

1

Raw Dataset

- patients.csv
- vitals_timeseries.csv
- labs_timeseries.csv
- hospital_deterioration_hourly_panel.csv

2

Objective and Scope

- Predict Decline within 12 hours
- Enable proactive intervention
- 72 Hours window for 10k patients

3

Methodological Rigor and EDA

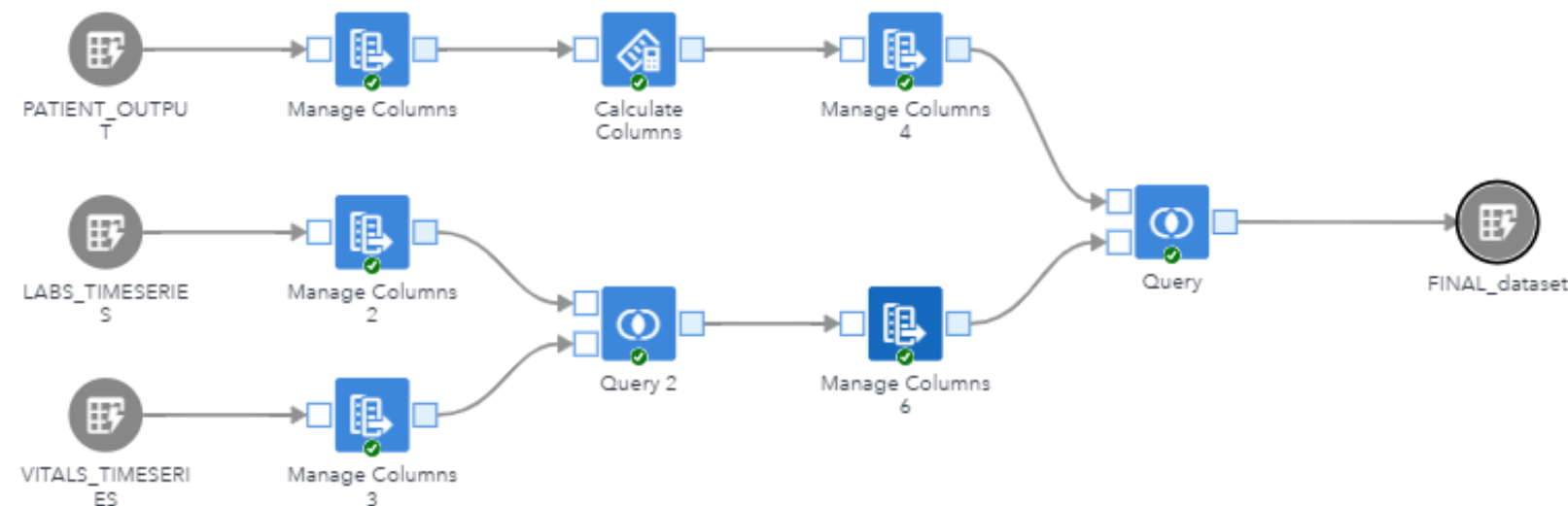
- Merge data streams
- Identify critical vital trends
- Assess risk feature impact

4

Visual Analytics and Storytelling

- Interactive Clinical dashboards
- SAS Path Analysis
- Recommendations for decision-makers

Dataset Selection & Justification



Patients Information

- 10,000 unique patients
- Demographics and risk
- One row per admission

Vitals Information

- 720,000 hourly rows
- Real-time vital signs
- Continuous patient monitoring

Labs Information

- Hourly blood markers
- Sepsis risk scores
- 72-hour clinical data

Final_dataset

- Unified hourly records
- Merged clinical features
- ML-ready target labels

How large is the deterioration problem and how is the population distributed across monitoring priority groups?

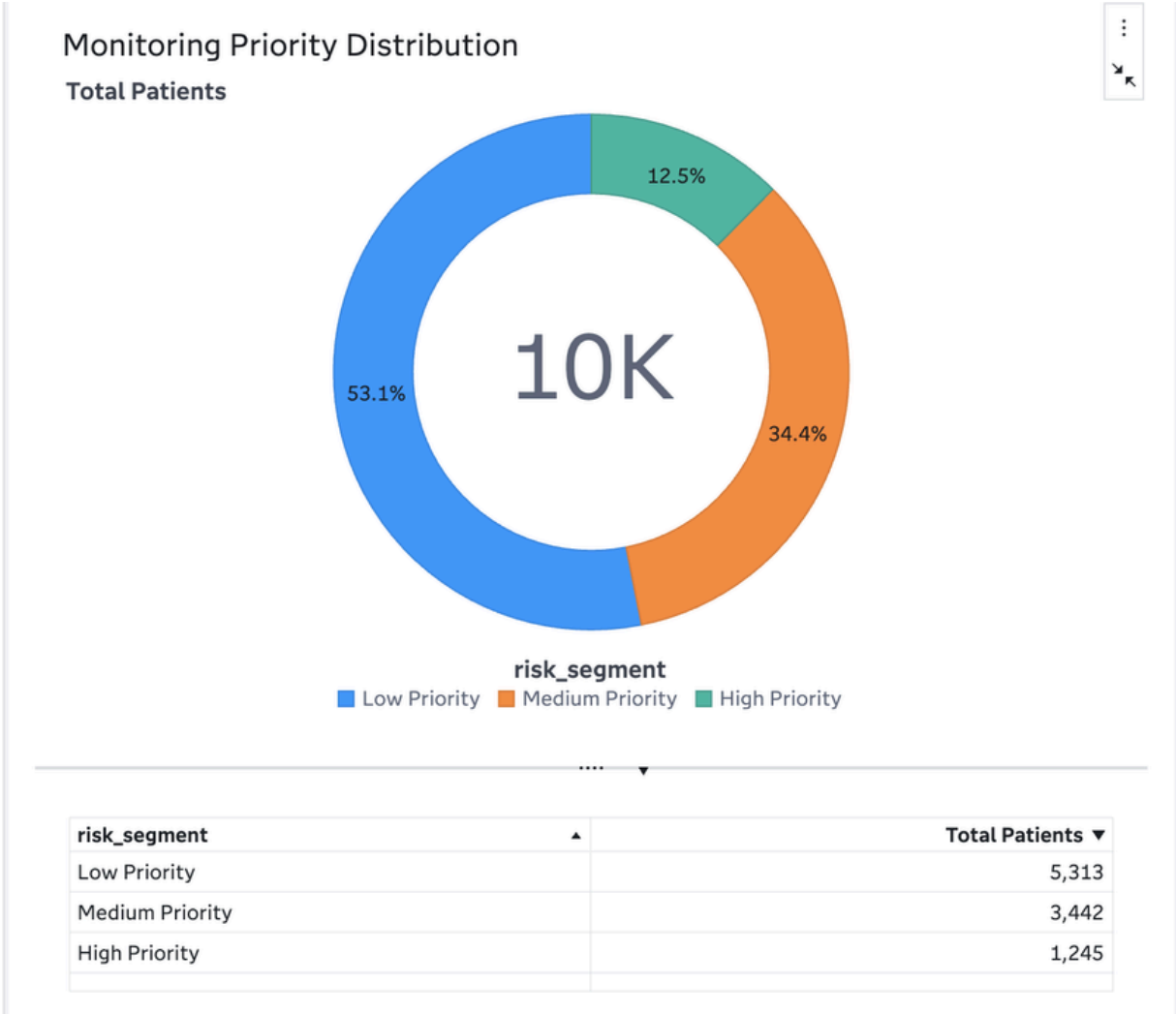
Total Patients
10K

Hourly Observations
418K

Patients Deteriorating
19%

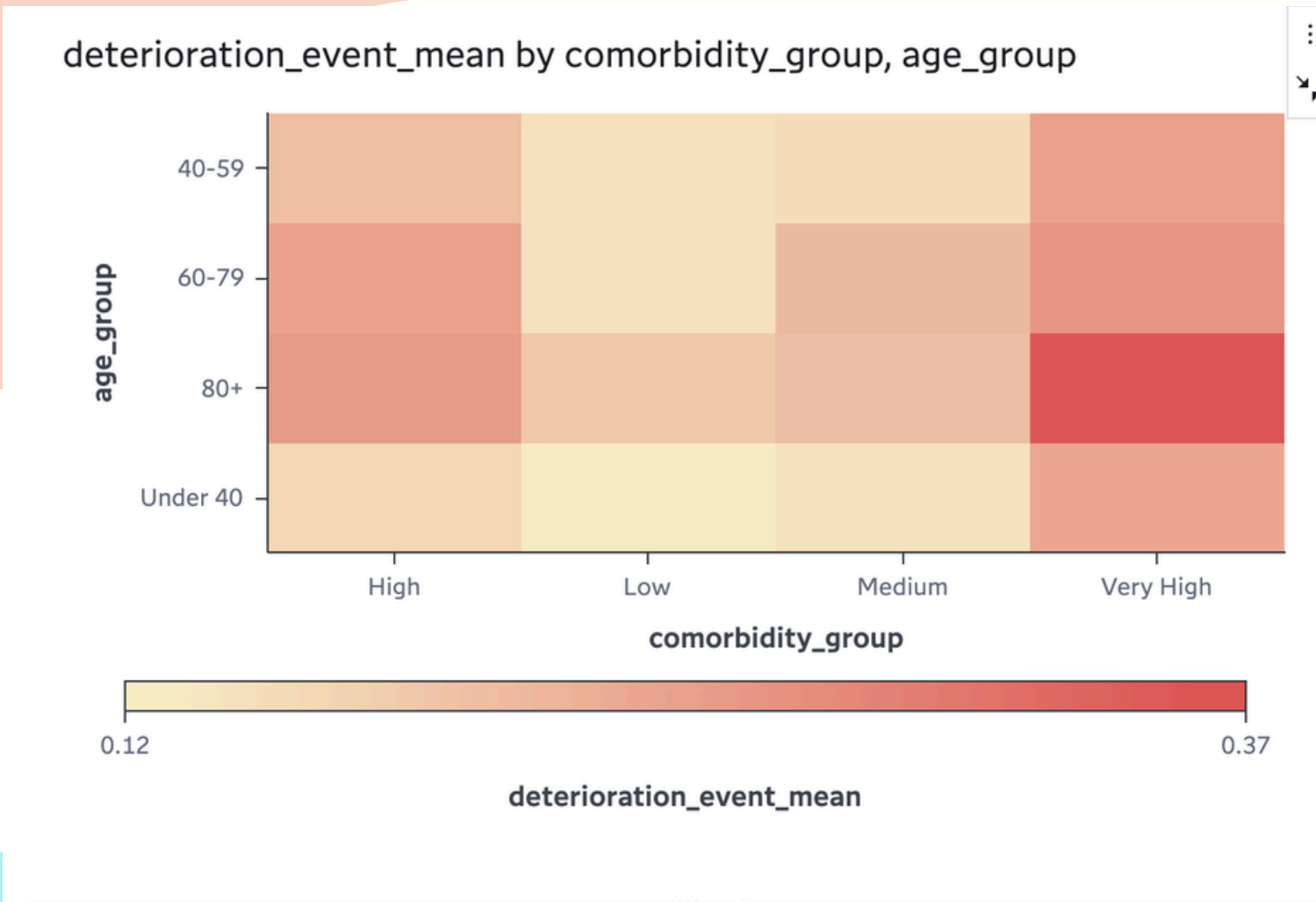
Near-Term Risk
3.1%

Nearly 1 in 5 patients deteriorated, showing a clear need for structured risk stratification and early monitoring.



Only 12.5% of patients are High Priority, making targeted enhanced monitoring operationally realistic.

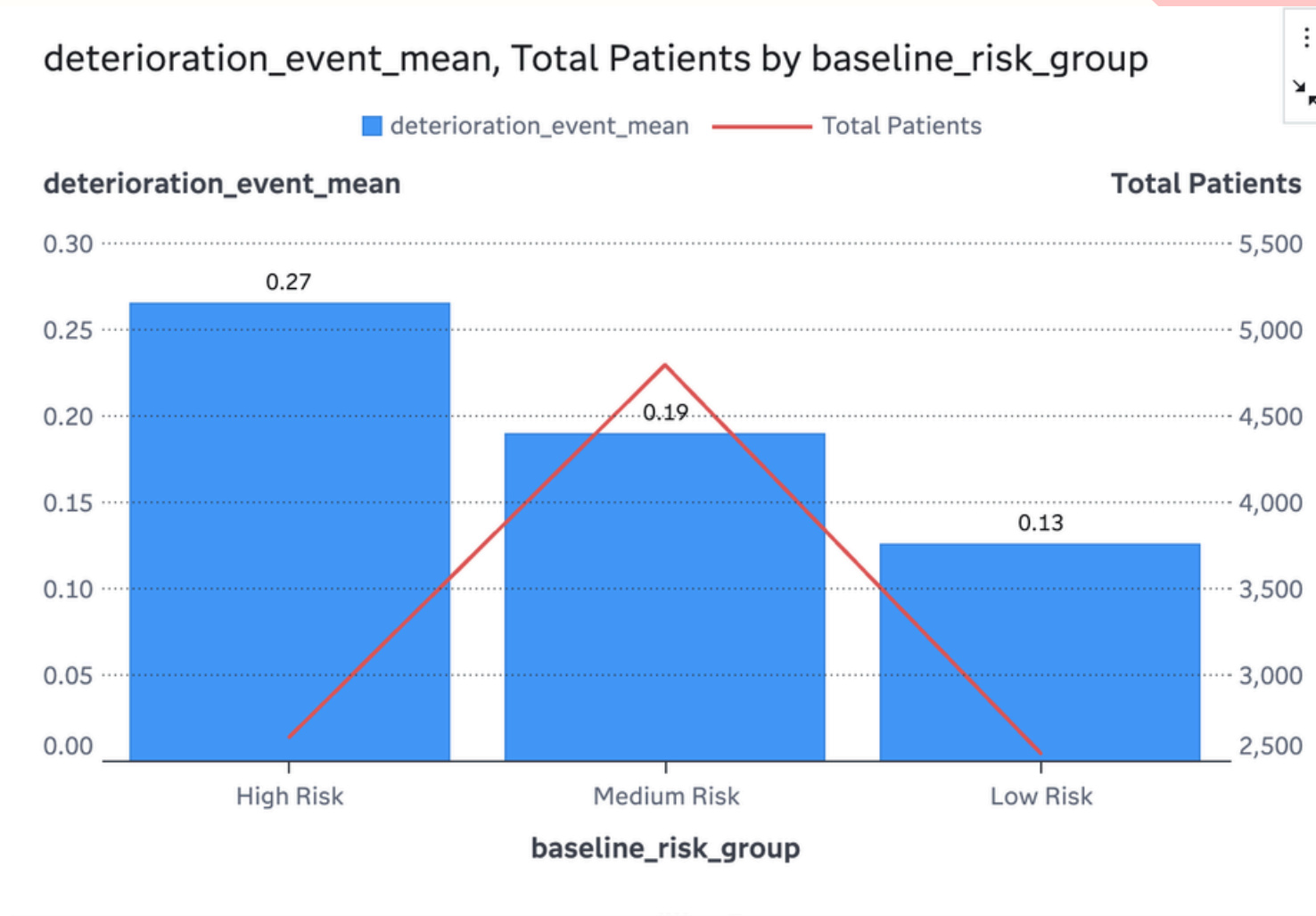
Which patient groups deteriorate most based on age and comorbidity?



comorbidity_group	age_group	deterioration_event_mean
High	40-59	0.1961823966
Low	40-59	0.1362889984
Medium	40-59	0.1444099379
Very High	40-59	0.2420212465

The highest-risk segment is 80+ patients with Very High comorbidity, reaching approximately 37% deterioration risk.

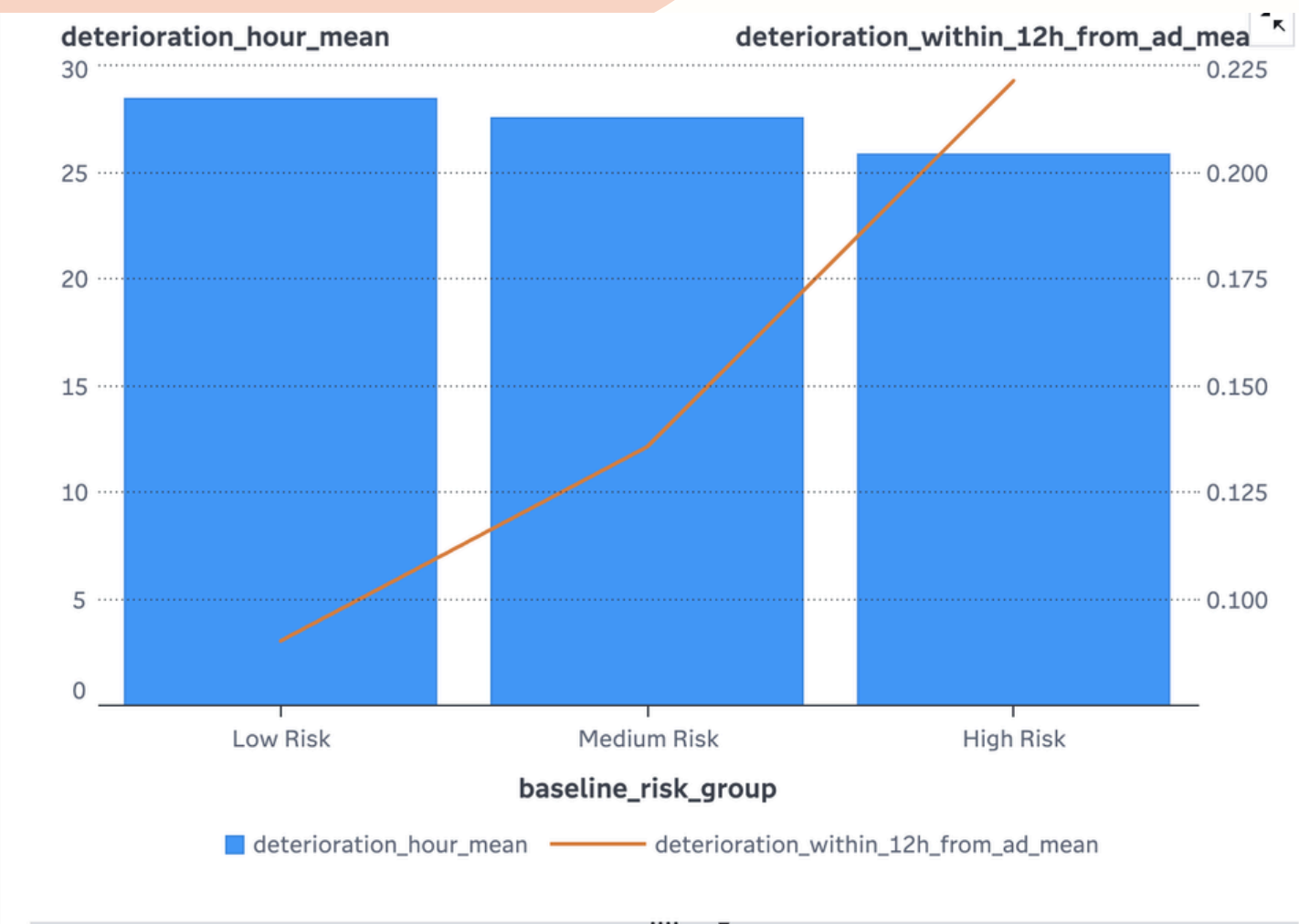
Does the baseline risk score effectively separate low-risk and high-risk patients?



baseline_risk_group	deterioration_event_mean	Total Patients
High Risk	0.2660619803	2,646
Medium Risk	0.1901291129	4,802
Low Risk	0.1257836991	2,552

High Risk patients deteriorate more than twice as often as Low Risk patients, showing that the baseline score is useful for prioritization.

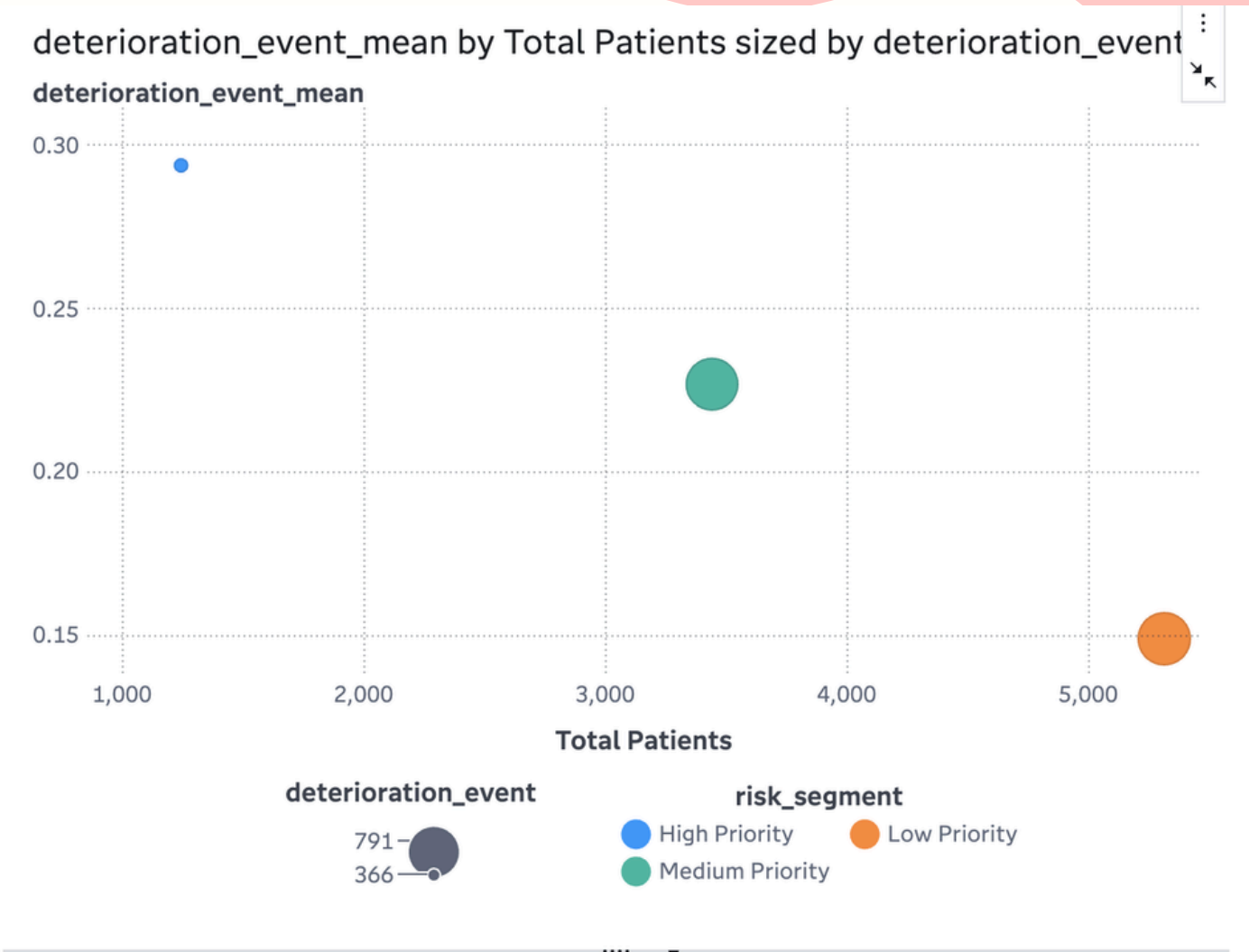
Do high-risk patients deteriorate earlier and need faster intervention?



baseline_risk_group	deterioration_hour_mean	deterioration_within_12h_from_a d_mean
Low Risk	28.473520249	0.0903426791
Medium Risk	27.570646221	0.1358159912
High Risk	25.865056818	0.2215909091

High Risk patients deteriorate earlier and are more likely to deteriorate within 12 hours, supporting faster clinical review.

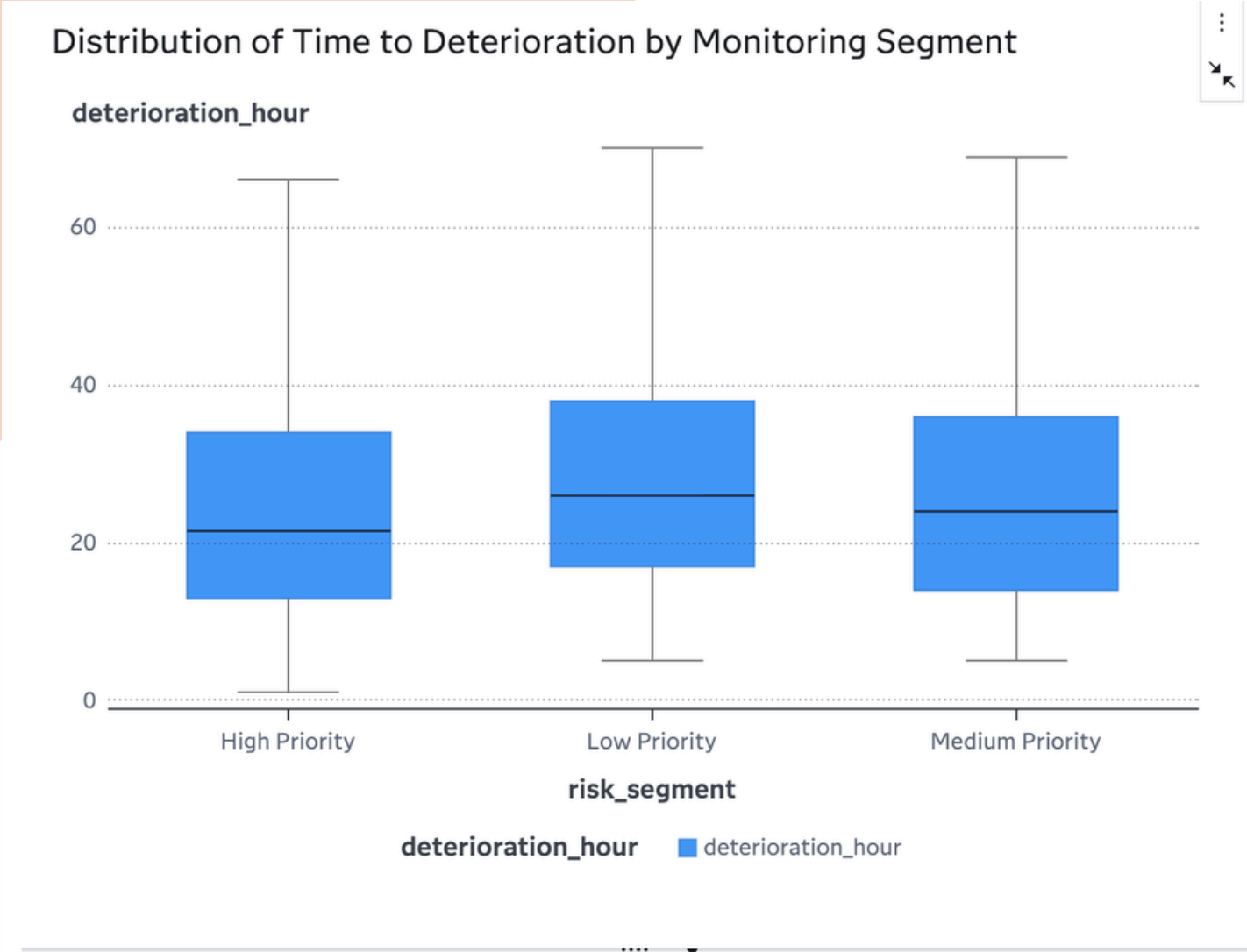
Which monitoring priority segment creates the greatest combination of risk and workload?



Total Patients	deterioration_event_mean	deterioration_event	risk_segment
1,245	0.2939759036	366	High Priority
5,313	0.1488801054	791	Low Priority
3,442	0.2269029634	781	Medium Priority

High Priority patients are the riskiest, but Low and Medium Priority groups still create major workload because of their larger size.

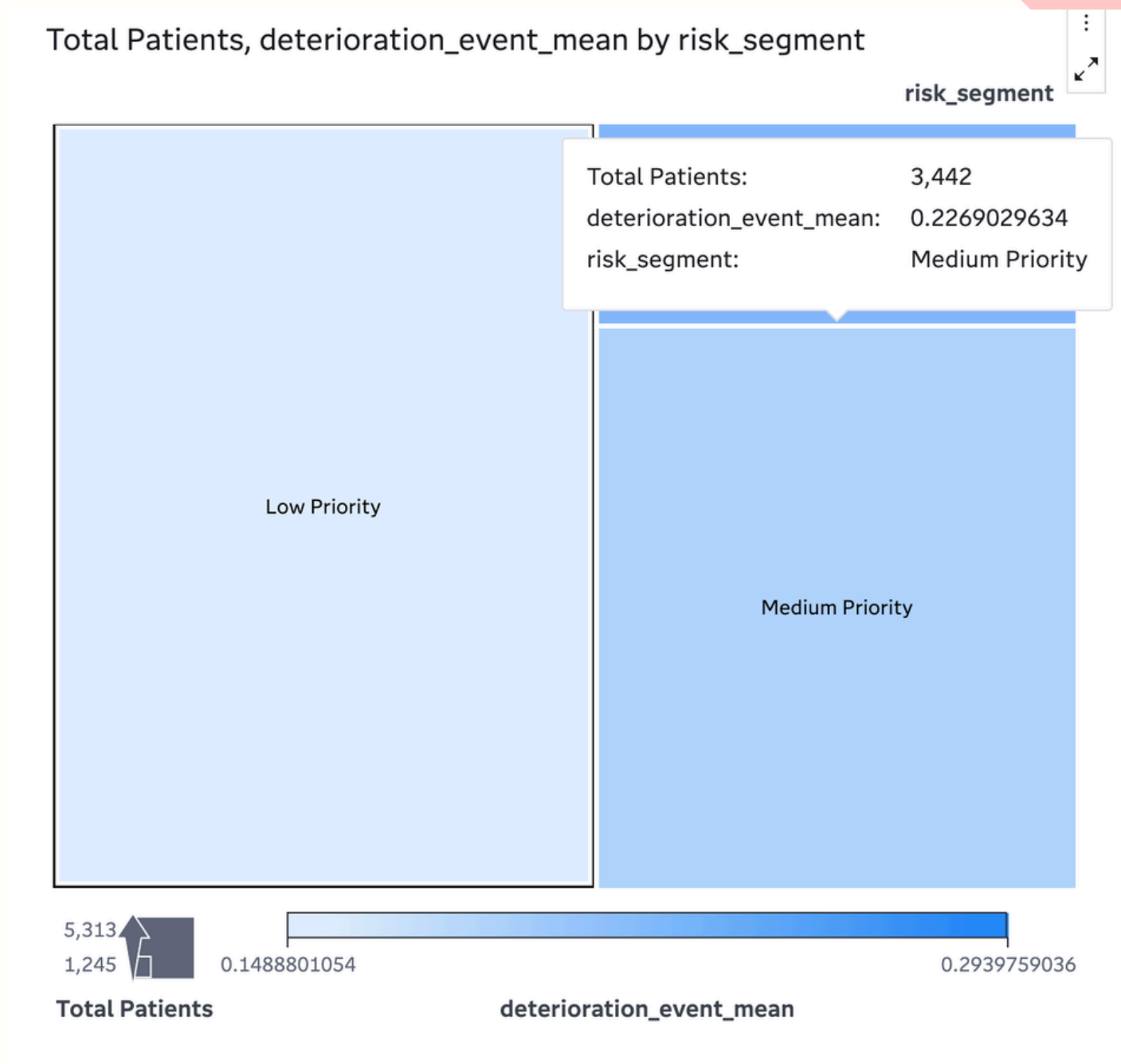
Do monitoring priority segments differ in how quickly patients deteriorate?



risk_segment ▲	Minimum	Lower Whisker	First Quartile	Average	Median	Third Quartile	Upp
High Priority	1	1	13	24.953551913	21.5	34	
Low Priority	5	5	17	28.579013906	26	38	
Medium Priority	5	5	14	26.609475032	24	36	

High Priority patients deteriorate around 4.5 hours earlier than Low Priority patients by median time, supporting faster intervention.

Where is patient volume and deterioration risk concentrated across monitoring priority segments?



The treemap shows both workload and risk. Low Priority is the largest group, but High Priority has the highest deterioration rate. Medium Priority is important because it combines a large patient volume with elevated risk.

Which factors increase the risk of patient deterioration?

patient_id
39K

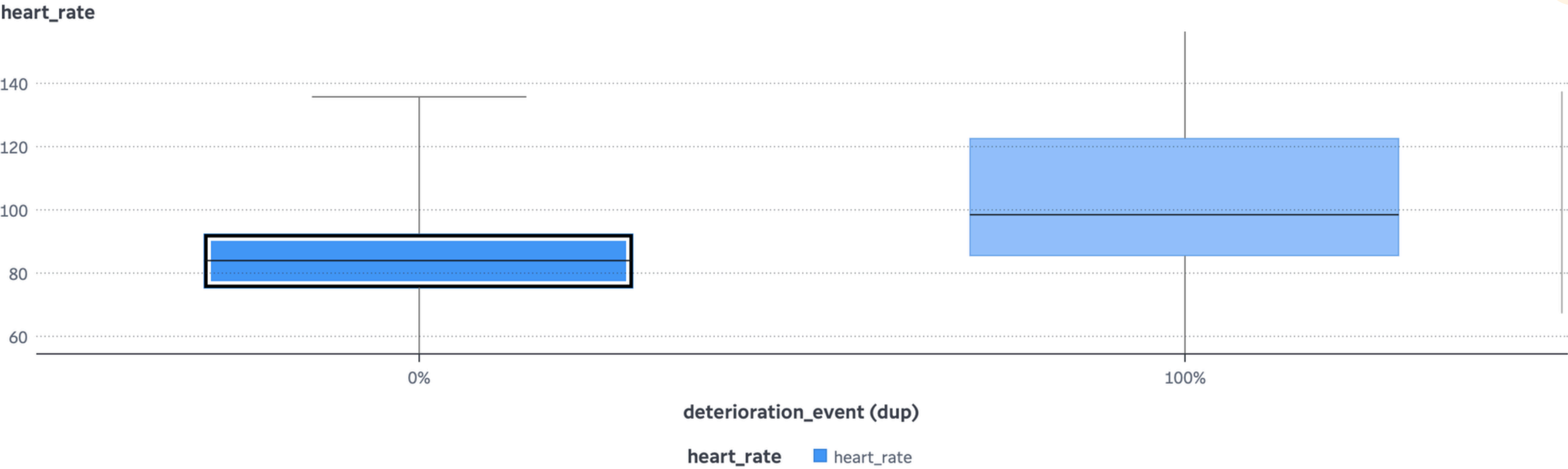
sepsis_risk_score
0.49

deterioration_event
22%

Used to compare heart rate distribution between stable and deteriorating patients.

Key Findings
Deteriorating patients had significantly higher heart rates
Average heart rate increased from approximately 84 BPM to 108 BPM

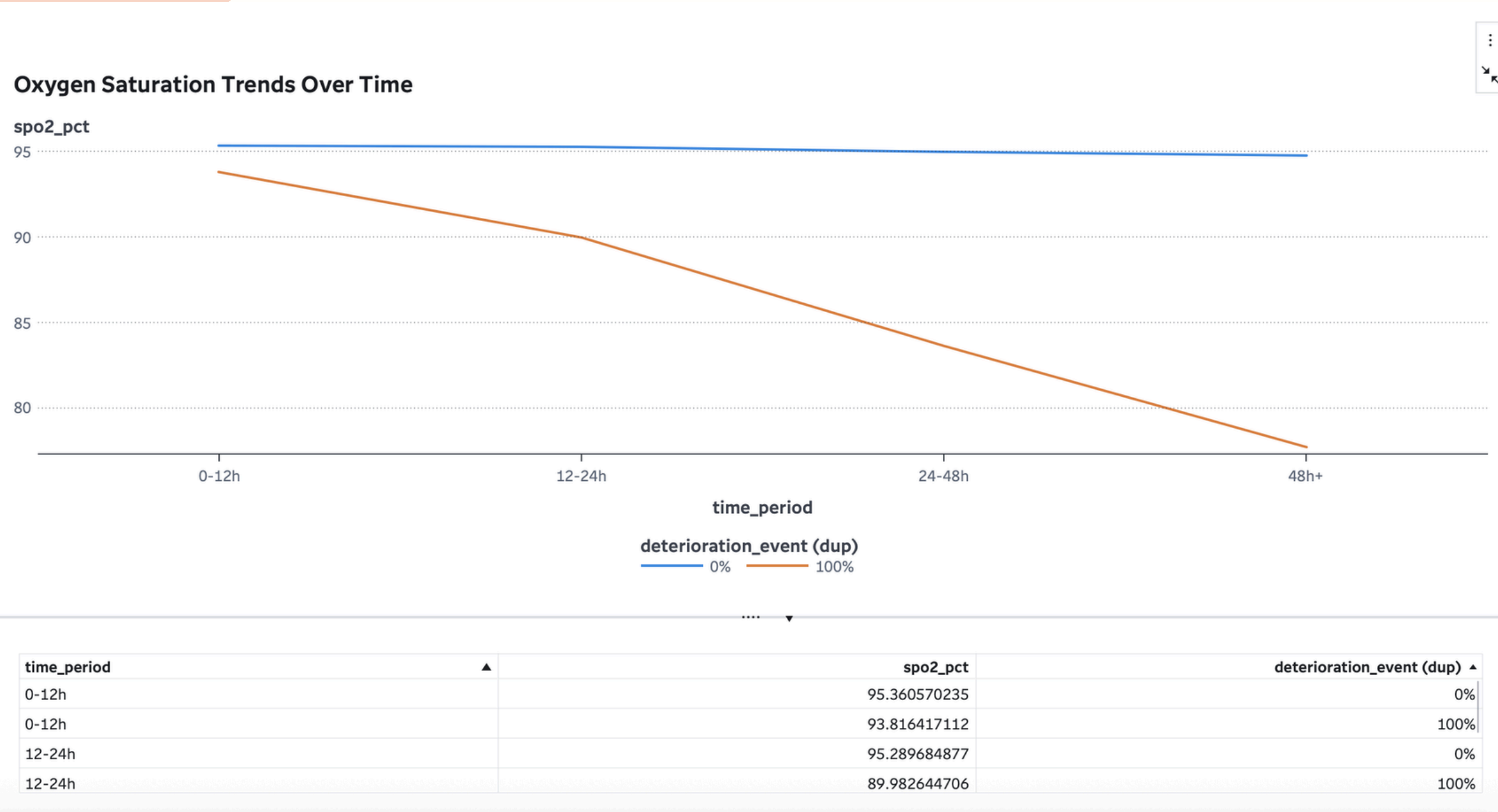
Heart Rate by Deterioration Status



deterioration_event (d... ▲	Minimum	Lower Whisker	First Quartile	Average	Median	Third Quartile	Upper Whisker	Maximum	Std Dev	Count
0%	40	40	75.61	83.970244625	84.1	92.65	135.74	135.74	13.065649141	30,373
100%	44.85	44.85	85.59	108.01294504	98.57	122.7	180	180	31.293347222	8,533

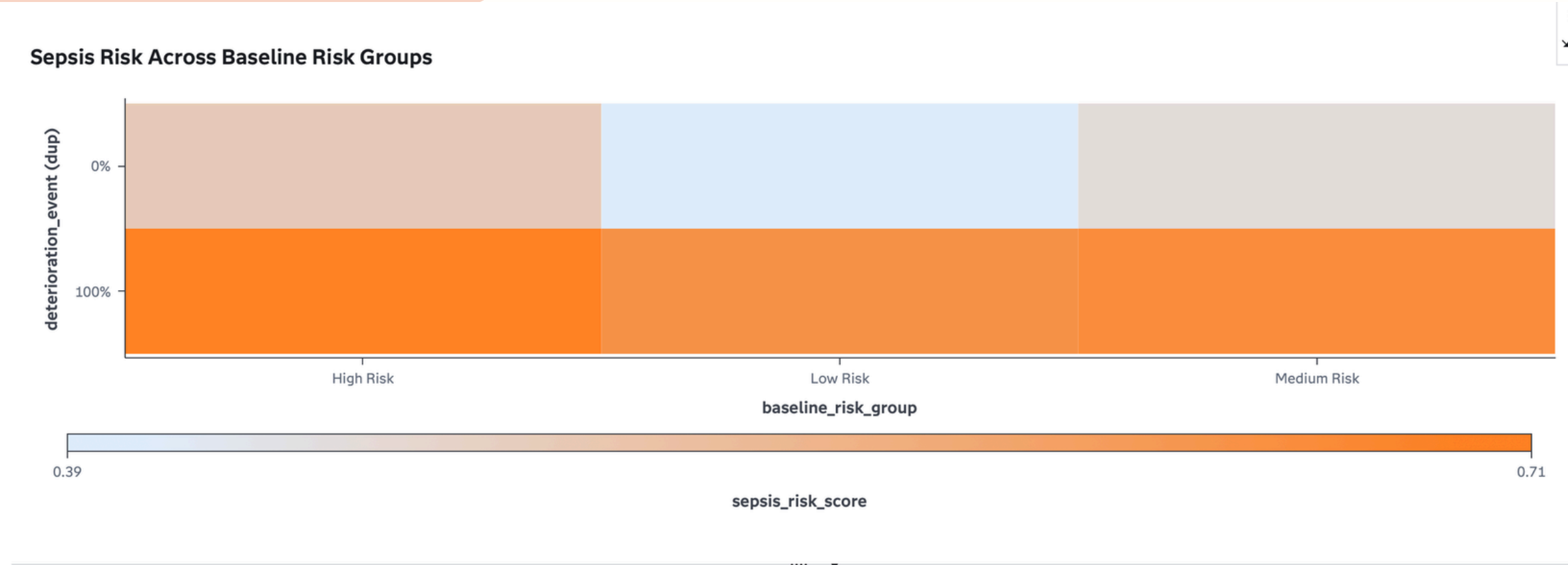
Which factors increase the risk of patient deterioration?

Used to analyze oxygen saturation trends over time.



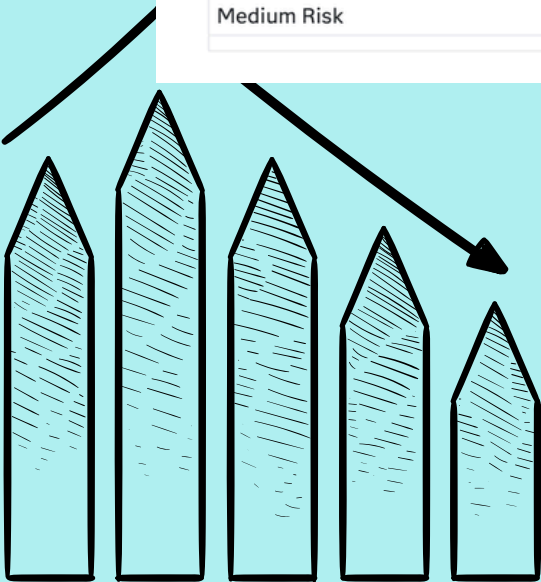
- Key Findings
- Oxygen saturation continuously declined in deteriorating patients
 - Stable patients maintained oxygen saturation around 95%
 - Deteriorating patients dropped to approximately 78% SpO₂ after 48 hours

Which factors increase the risk of patient deterioration?



Key Findings

- High-risk patients showed the highest sepsis risk scores



Which factors increase the risk of patient deterioration?

Used to compare deterioration contribution across risk groups.

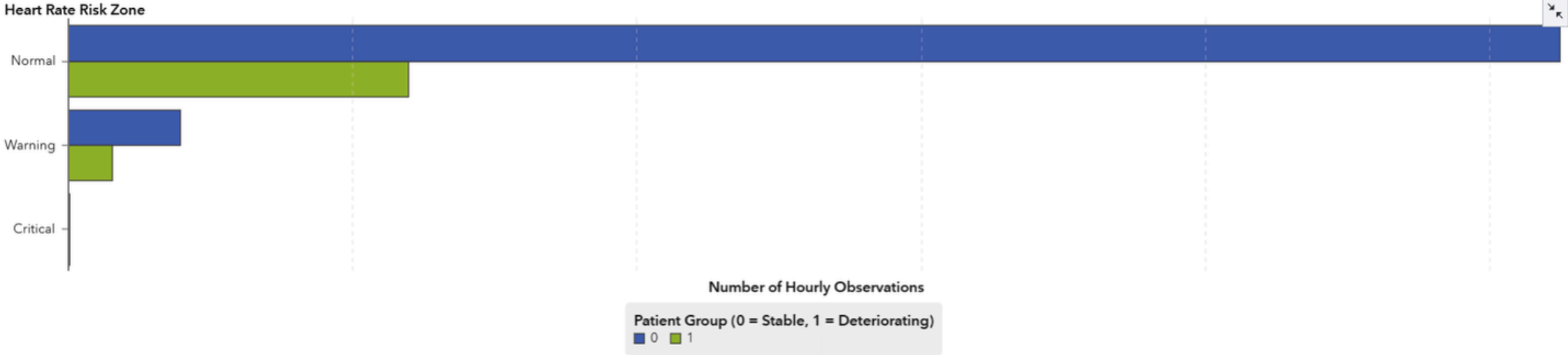


Key Findings

- High Risk group had approximately 32% deterioration rate
- Low Risk group had approximately 12% deterioration rate
- Higher sepsis risk strongly correlated with deterioration

What are the critical thresholds for heart rate?

Frequency of HR_Category grouped by deterioration_event

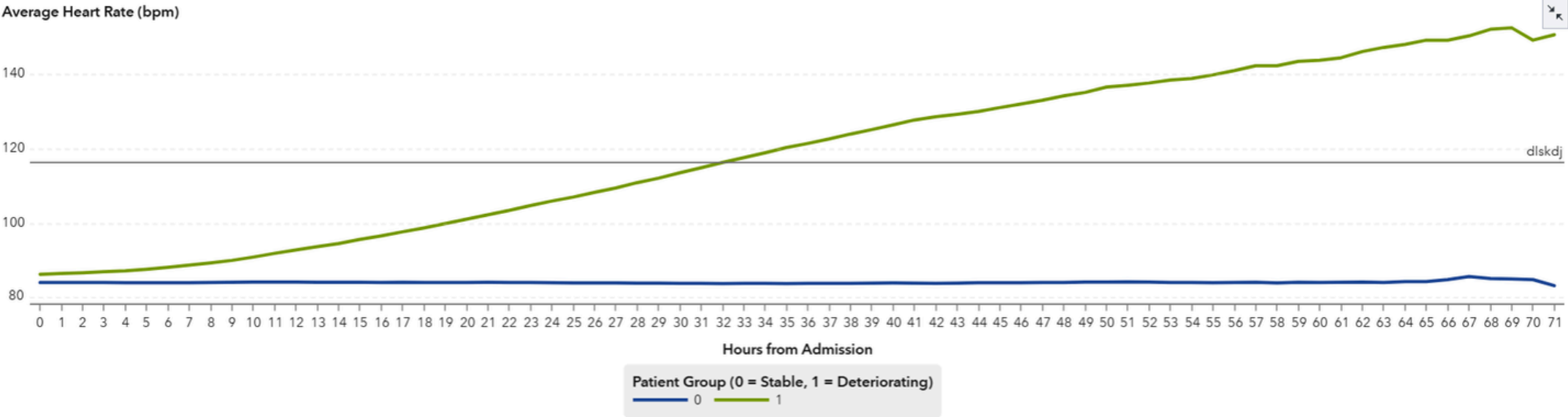


Majority of deteriorating patients fall in the Normal HR zone, Heart rate alone is not enough to detect deterioration

Heart Rate Risk Zone	Number of Hourly Observations	Patient Group (0 = Stable, 1 = Deteriorating)
Normal	52,438	0
Normal	11,987	1
Warning	3,968	0
Warning	1,577	1
Critical	28	0
Critical	2	1

Deteriorating patients show continuously rising Heart Rate from hour zero to hour 71, While stable patients remain flat showing deterioration begins at admission

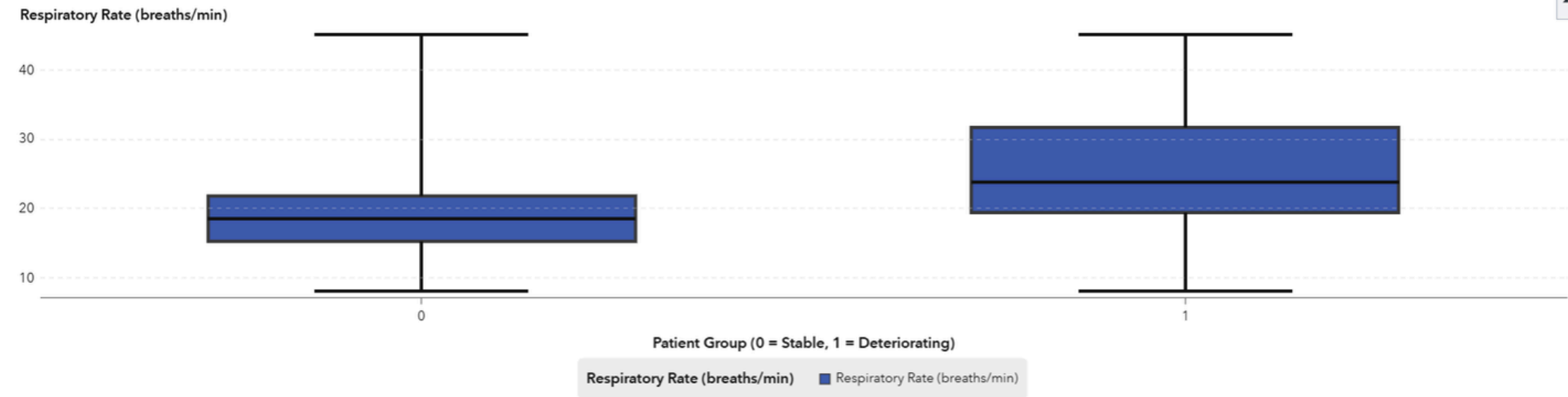
Average Heart Rate Trajectory by Deterioration Status



Hours from Admission	Average Heart Rate (bpm)	Patient Group (0 = Stable, 1 = Deteriorating)
0	84.197773505	0
0	86.400092879	1
1	84.183130737	0
1	86.626186791	1

Which vital sign best predicts deterioration?

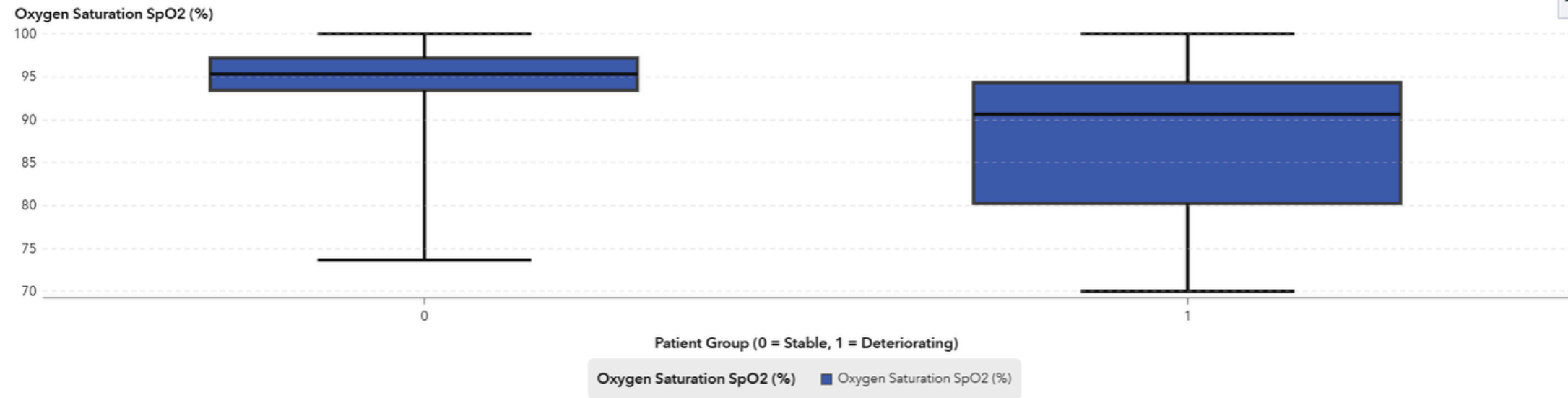
Respiratory Rate by Deterioration Status



Deteriorating patients breathe 8 times more per minute than stable patients . Respiratory Rate is the strongest predictor of deterioration

Deteriorating patients have 7% lower oxygen saturation . SpO2 is the second strongest predictor after Respiratory Rate

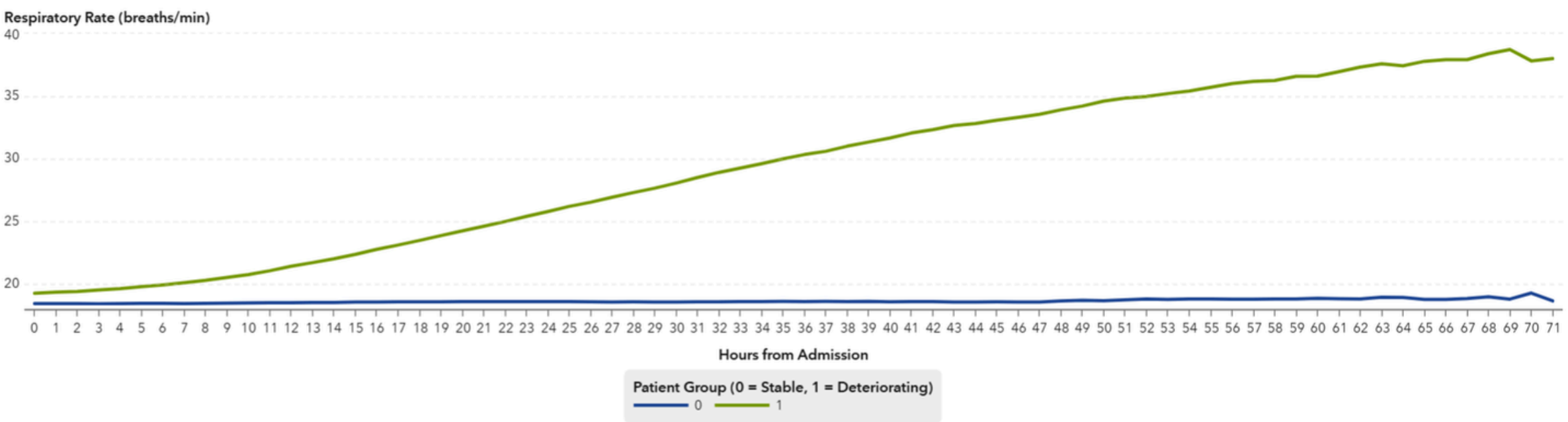
SpO2 by Deterioration Status



Patient Group (0 = Stable, 1 = Deteriorat... ^	Minimum	Lower Whisker	First Quartile	Average	Median	Third Quartile	Upper Whisker	Maximum	Std Dev	Count
0	73.64	73.64	93.5	95.233309456	95.37	97.24	100	100	2.7344953175	3,30,027
1	70	70	80.34	87.011037352	90.67	94.32	100	100	9.3245284118	87,839

Is deterioration sudden or gradual?

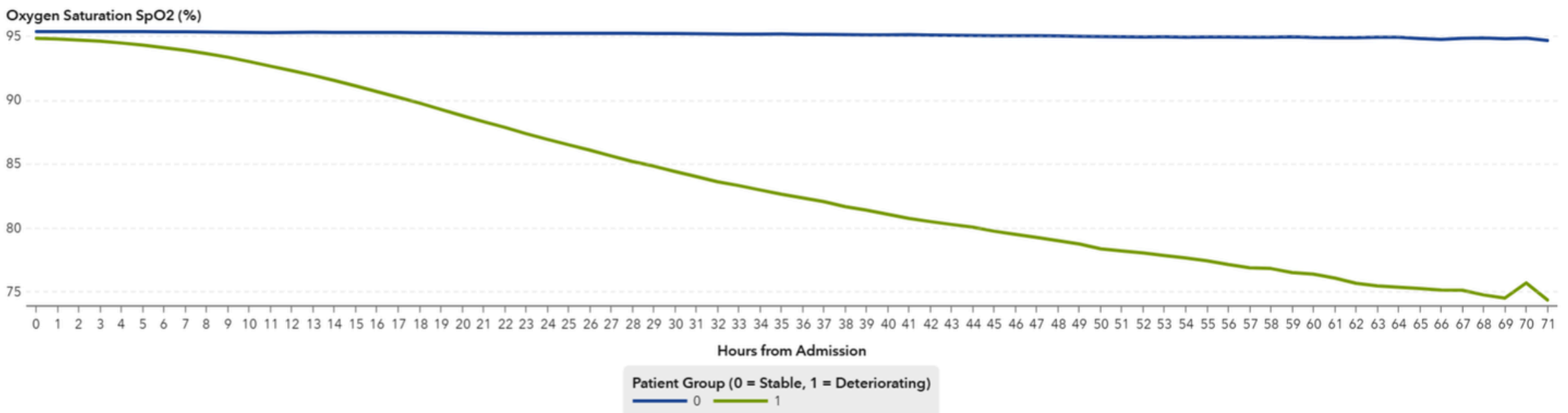
Respiratory Rate Trajectory by Deterioration Status



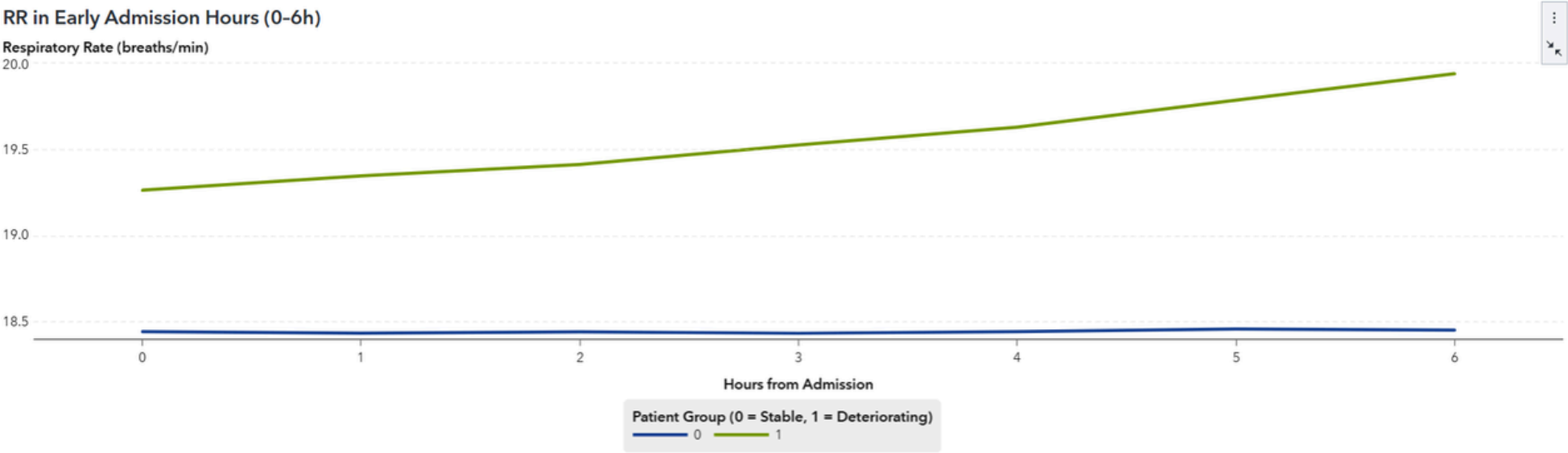
Respiratory Rate rises continuously from 20 to 38 breaths/min over 72 hours in deteriorating patients , while stable patients stay completely flat at 19 , proving deterioration is GRADUAL not sudden

Oxygen Saturation drops steadily from 94% to 74% over 72 hours in deteriorating patients , while stable patients remain flat at 95% , both vitals tell the exact same deterioration story from opposite directions

SpO2 Trajectory by Deterioration Status

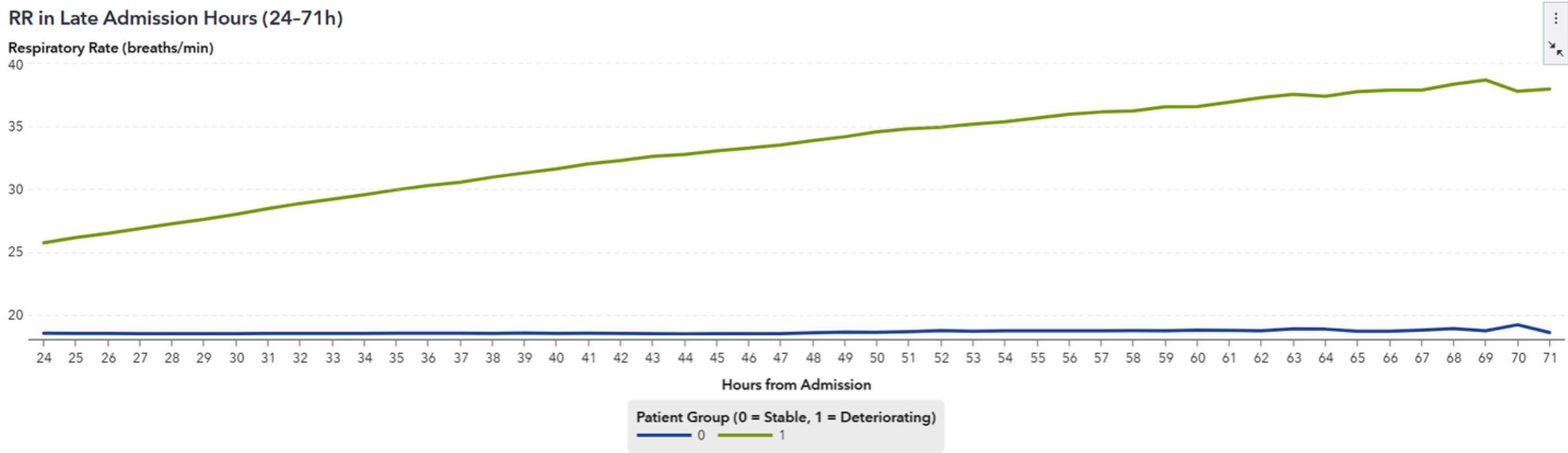


Are early warning signs different from late ones?



In the first 6 hours the gap between deteriorating and stable patients is only 1.5 breaths/min , so small it is almost impossible to detect with standard monitoring

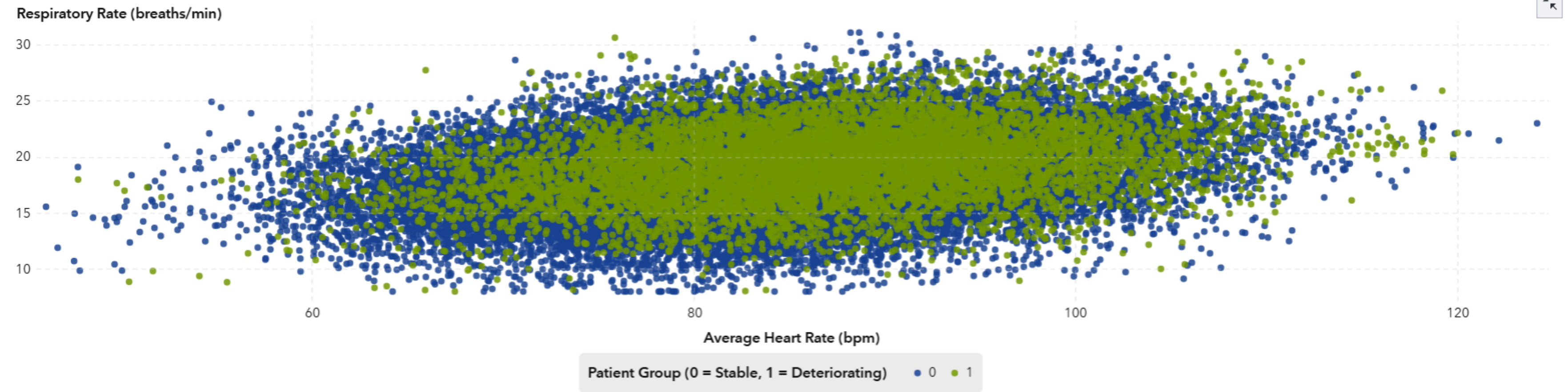
By hour 24 the gap grows to 15 breaths/min , ten times larger — deterioration is now clearly visible but precious time has already been lost



Hours from Admission	Respiratory Rate (breaths/min)	Patient Group (0 = Stable, 1 = Deteriorating)
24	18.606591943	0
24	25.787043427	1
25	18.594124228	0
25	26.203054545	1

Does combining two vitals improve risk prediction?

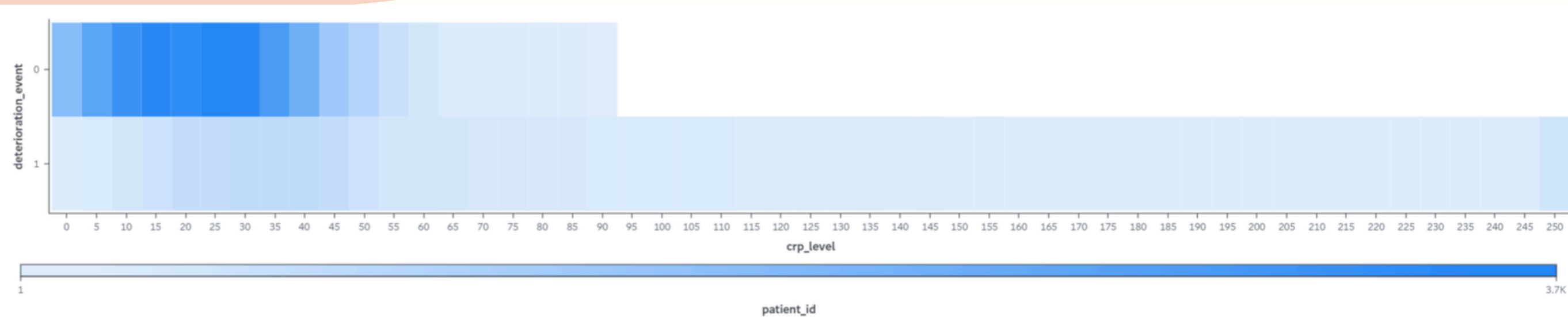
Combined HR & RR Risk – Stable vs Deteriorating Patients



Average Heart Rate (bpm)	Respiratory Rate (breaths/min)	Patient Group (0 = Stable, 1 = Deteriorating)
72.97	17.07	0
81.6	16.24	0
80.15	10.76	0
84.3	15.82	0

Patients with BOTH high Heart Rate AND high Respiratory Rate at the same time are more likely to deteriorate — two warning signs together are stronger than one alone

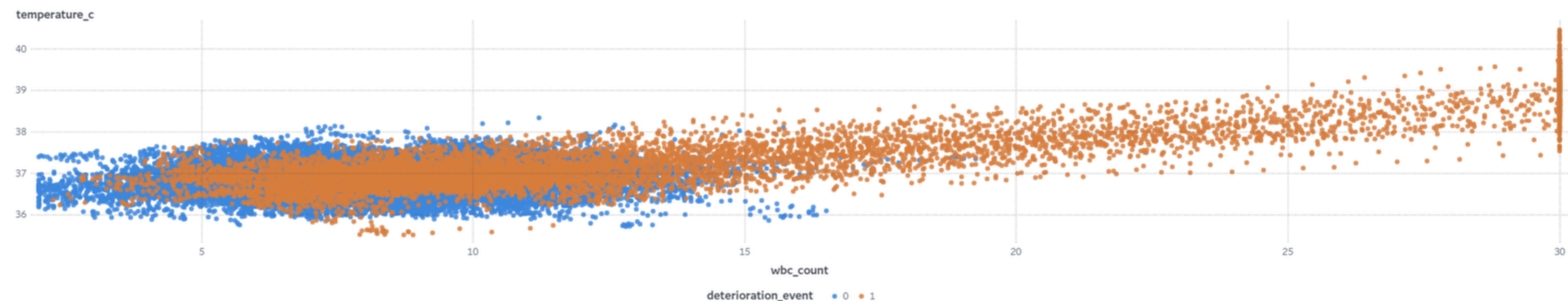
Do inflammation markers increase in deteriorating patients?



Elevated CRP ranges appear among deteriorating patients,

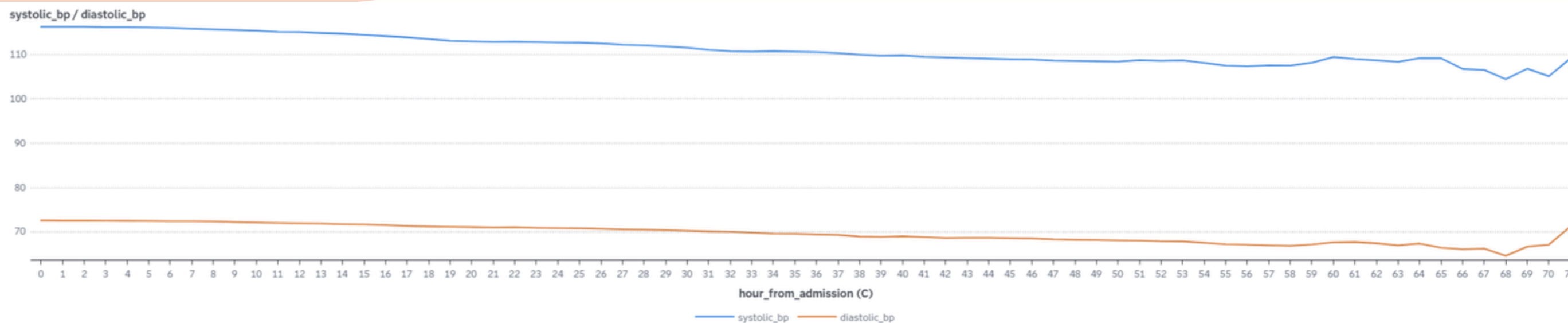
Are infection indicators associated with deterioration?

infection-related biomarkers may be strongly associated with patient deterioration.



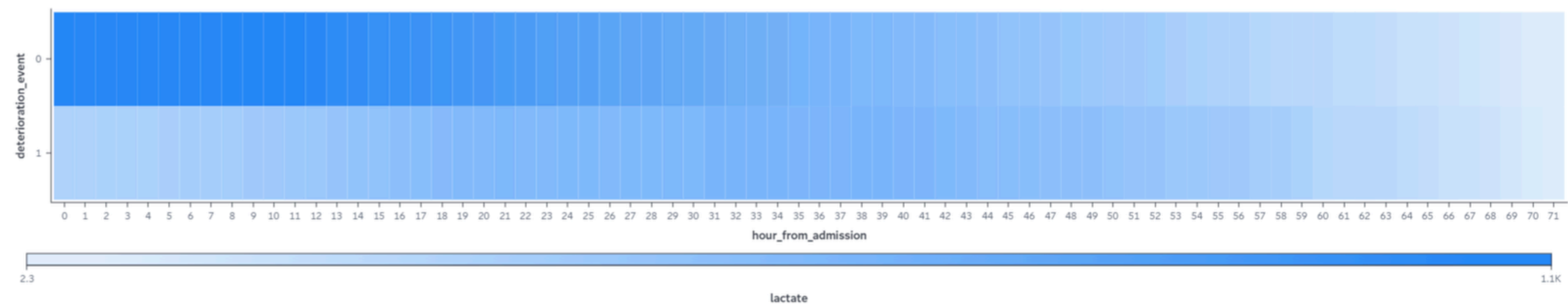
How do blood pressure patterns change over hospitalization before deterioration?

this pattern may indicate blood pressure fluctuations over the duration of stay.



Do lactate levels indicate metabolic stress before deterioration?

The decreasing lactate over time may indicate successful physiological stabilization.



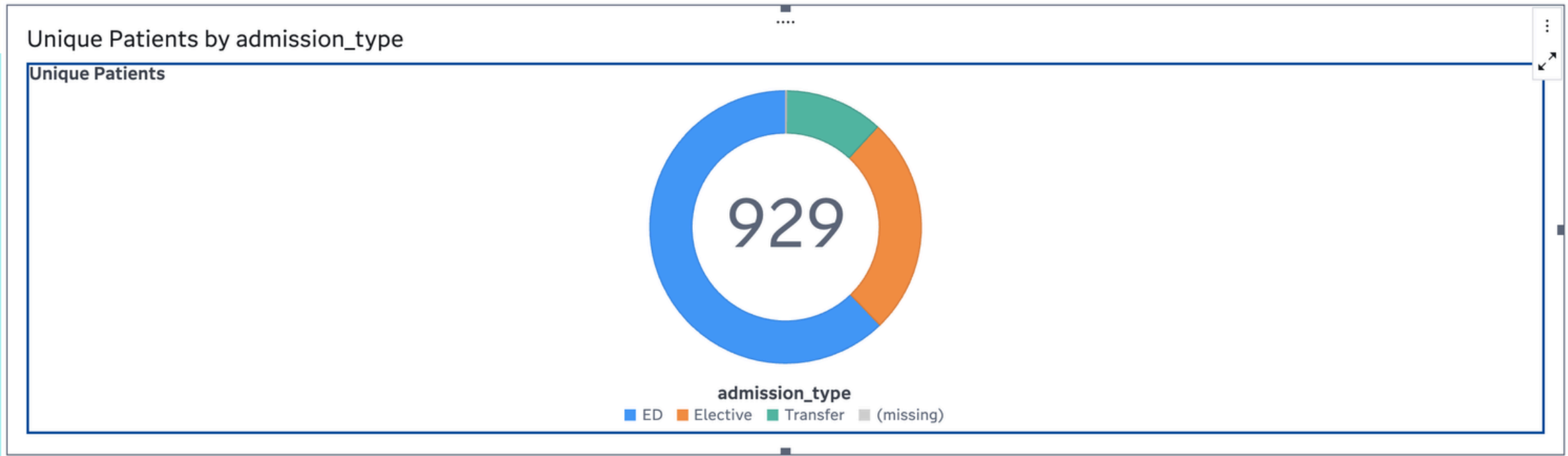
Do kidney function abnormalities relate to deterioration?



Key Findings

- Creatinine Spikes: Elevated levels are significantly more common in deteriorating patients.
- Kidney Stress: High readings serve as a direct marker of worsening organ function.
- Systemic Instability: Reflects general bodily failure beyond a single organ.
- Clinical Warning: Abnormal distributions identify patients at high risk of instability.

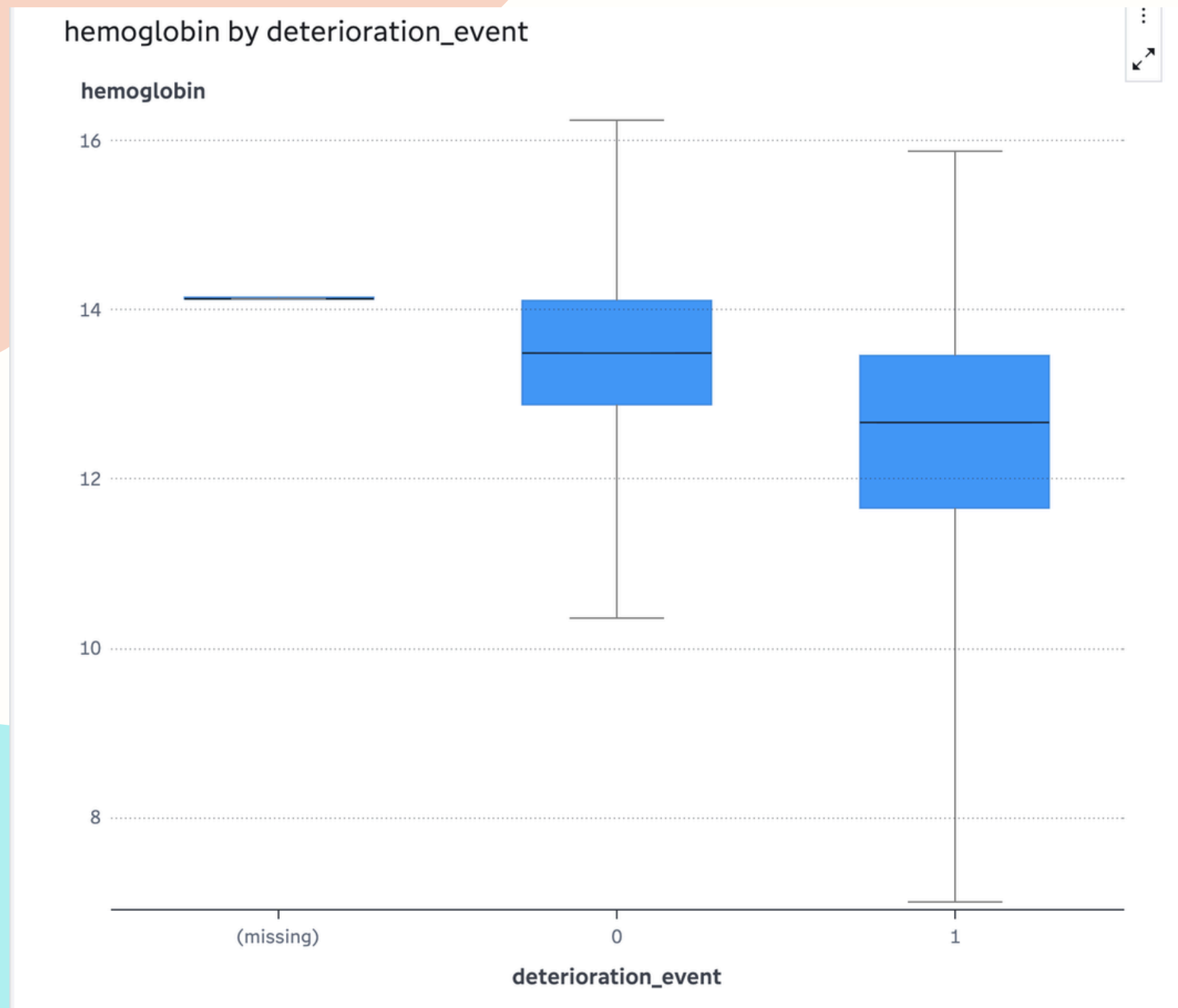
Are transferred patients more clinically vulnerable?



Key Findings

- Increased Risk: Transferred patients show higher rates of clinical deterioration.
- Patient Fragility: Transfer status signals higher medical vulnerability and instability.
- Care Complexity: Reflects the advanced needs and difficulty of managing these cases.
- Priority Oversight: Highlights the need for closer monitoring for all transferred patients.

Does abnormal hemoglobin relate to deterioration risk?



Key Findings

- Holistic Monitoring: Look beyond single vitals to capture complex decline.
- Integrated Predictors: Combine risk scores, mobility, and lab values (\$Hb\$).
- Proactive Prevention: Shift from reactive rescue to data-driven intervention.
- System Impact: Improve patient safety and hospital resource efficiency.

Business Insights: From Patient Data to Monitoring Priorities

1. Deterioration is operationally important

Nearly 1 in 5 patients deteriorated, showing that early monitoring is a relevant hospital safety issue.

2. Risk is not evenly distributed

Patients with high baseline risk, older age, and very high comorbidity show the highest deterioration risk.

3. Baseline risk score is useful

Actual deterioration increases from 12.6% in Low Risk patients to 26.6% in High Risk patients.

4. Risk segmentation supports targeted monitoring

Only 12.5% of patients are High Priority, but they have the highest deterioration rate at 29.4%.

5. Timing matters

High Priority patients deteriorate earlier, with a median deterioration time of 21.5 hours compared with 26 hours for Low Priority patients.

6. Single indicators are not enough

Vitals, labs, oxygen support, and patient profile should be combined into one monitoring workflow.



Recommendations for Decision-Makers



1. Use risk segmentation at admission

Classify patients into Low, Medium, and High Priority groups using age, comorbidity, admission type, and baseline risk score.

2. Prioritize High Priority patients in the first 24 hours

High Priority patients deteriorate earlier and should receive faster clinical review after admission.

3. Combine patient profile with live clinical signals

Use baseline risk together with respiratory rate, SpO₂, heart rate, lactate, creatinine, oxygen support, and nurse alerts.

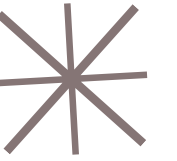
4. Build an early-warning dashboard

Show each patient's risk segment, warning signs, deterioration risk, and time since admission.

5. Support staff allocation

Use the dashboard to focus monitoring resources on patients with the highest risk and highest urgency.





Thank You!

